



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions.

(6x5=30)

- How many pairs of observation must be included in the sample in order that an observed correlation coefficient of value 0.47 shall have a calculated value of t greater than 2.12.
- Given that

X_1	3	5	6	8
X_2	18	9	7	4
X_3	90	27	45	24

Find all partial correlation coefficients.

- Differentiate between multiple and partial correlations.
- Write the advantages and disadvantages of completely randomized Block design.
- The grades in a statistics course were as follow

Grades	A	B	C	D	E
f	14	18	32	20	16

Test the hypothesis that the Distribution of grades is uniform. Let $\alpha=0.05$

- Name the different multiple range tests. Explain any one of them.

Answer the following questions

(3x10=30)

Q2. The following data represent the marks obtained by five students in three subjects

Student	Subjects		
	English	Statistics	Economics
1	47	73	61
2	80	88	86
3	71	77	59
4	62	60	66
5	56	95	87

Use a 0.05 level of significance to test the hypothesis that

- The subjects are of equal difficulty.
- The students have equal ability.

Q3. Estimate the regression line from the following data of heights (X) and weights (Y) of 12 persons

Heights(X)	Weights(Y)
60	110, 135, 120
62	120, 140, 130, 135
64	150, 145
70	170, 185, 160

Test the hypothesis that height and weight are independent

Q4. Consider the following set of data

X_1	4	6	5	8	12	13	19
X_2	1	1	2	3	4	5	7
X_3	2	4	3	4	8	8	11

Find the estimated regression line $X_1 = b_1 + b_2X_2 + b_3X_3$ and $R_{1.23}$.